

Ejercicios progresivos del Tema 3

Simplifica utilizando las leyes de álgebra de Boole (y Karnaugh si quieres) los siguientes circuitos:

1. $A(A + B)$
2. $A(\bar{A} + AB)$
3. $BC + \bar{B}C$
4. $A(A + \bar{A}B)$
5. $\bar{A}\bar{B}C + \bar{A}BC + \bar{A}\bar{B}C$
6. $(A + \bar{B})(A + C)$
7. $\bar{A}B + \bar{A}B\bar{C} + \bar{A}BCD + \bar{A}B\bar{C}\bar{D}E$
8. $AB + \overline{ABC} + A$
9. $(A + \bar{A})(AB + AB\bar{C})$
10. $AB + (\bar{A} + \bar{B})C + AB$
11. $BD + B(D + E) + \bar{D}(D + F)$
12. $\bar{A}\bar{B}C + \overline{A + B + \bar{C}} + \bar{A}\bar{B}\bar{C}D$
13. $(B + BC)(B + \bar{B}C)(B + D)$
14. $ABC[AB + \bar{C}(BC + AC)]$
15. $ABCD + AB(\overline{CD}) + \overline{AB}CD$
16. $\overline{A + \bar{B}}$
17. $\overline{\bar{A}B}$
18. $\overline{(A + \bar{B})(\bar{C} + D)}$
19. $\overline{\bar{A}B(C + \bar{D})}$
20. $\overline{A + \bar{B} + C + \bar{D}} + \overline{ABCD}$
21. $\overline{\overline{(\bar{A} + B + C + D)} (\bar{A}\bar{B}\bar{C}\bar{D})}$
22. $\overline{A + \overline{\bar{B}C}} + \overline{CD} + \overline{\bar{B}C}$